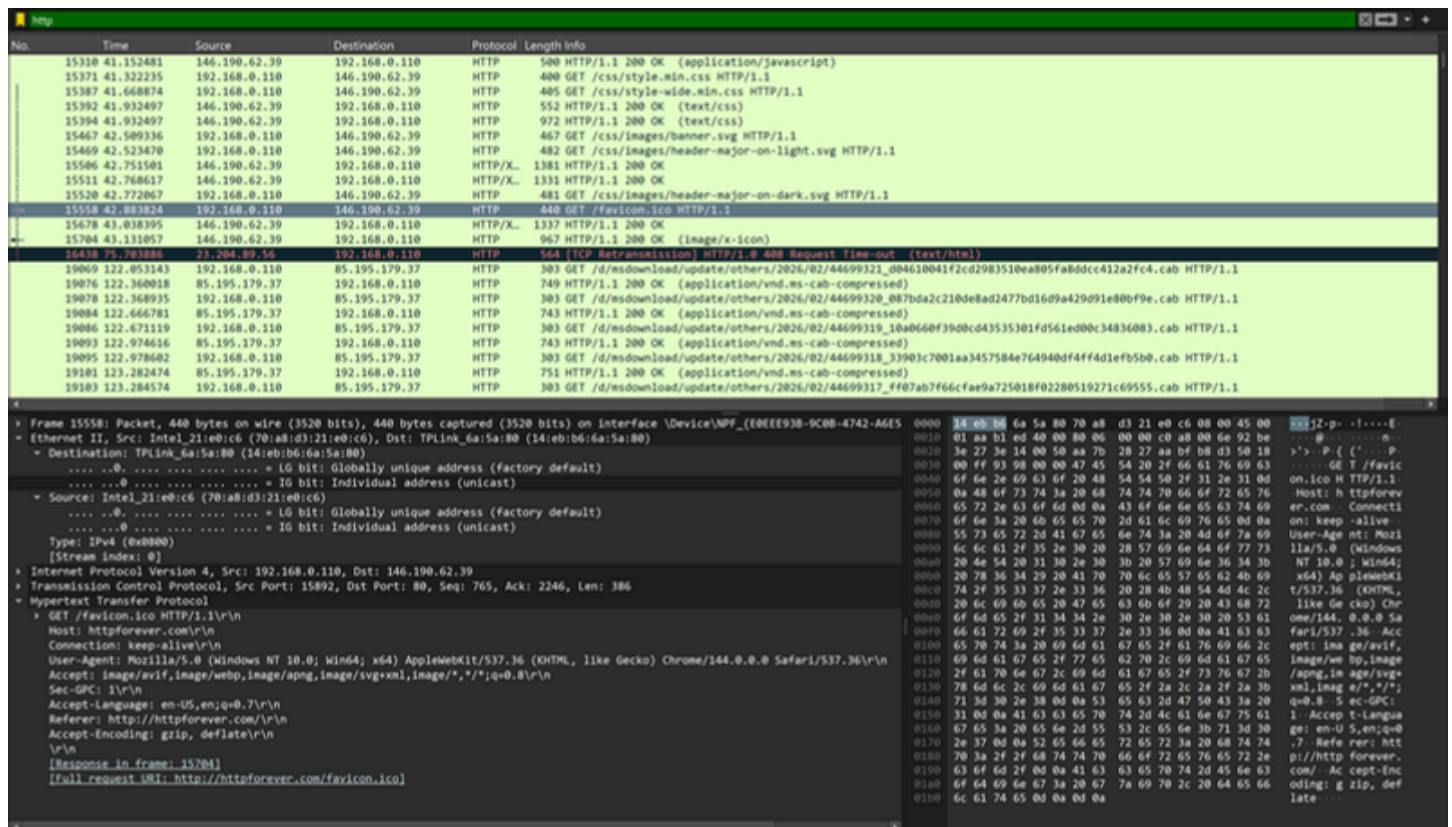


CS471-LAB1

لميس عبدالله الحربي - 431201519

Part 1: Capturing HTTP Traffic



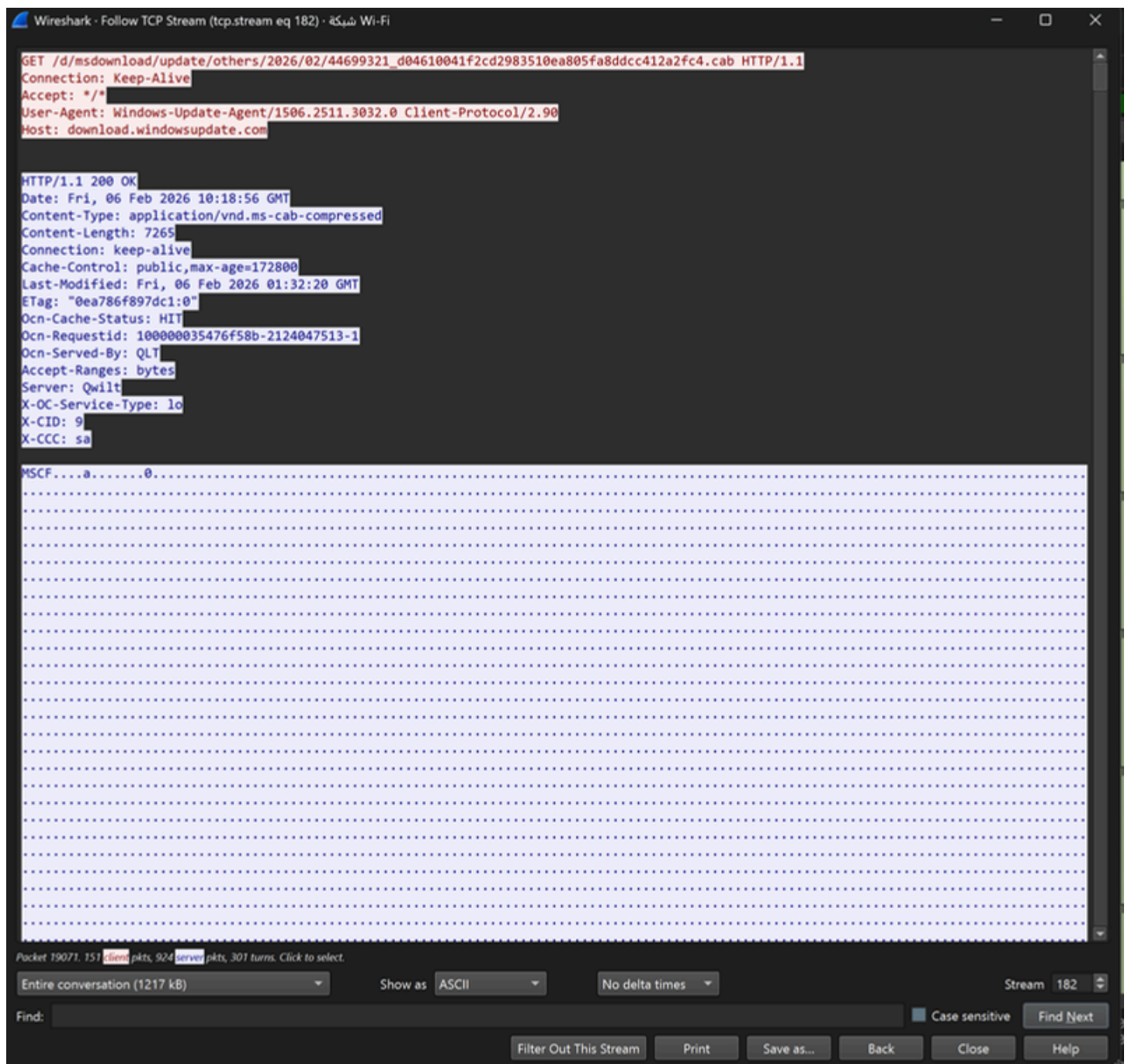
```
15310 41.152481 146.190.62.39 192.168.0.110 HTTP 500 HTTP/1.1 200 OK (application/javascript)
15371 41.322235 192.168.0.110 146.190.62.39 HTTP 400 GET /css/style.min.css HTTP/1.1
15387 41.668874 192.168.0.110 146.190.62.39 HTTP 405 GET /css/style-wide.min.css HTTP/1.1
15392 41.932497 146.190.62.39 192.168.0.110 HTTP 552 HTTP/1.1 200 OK (text/css)
15394 41.932497 146.190.62.39 192.168.0.110 HTTP 972 HTTP/1.1 200 OK (text/css)
15467 42.509336 192.168.0.110 146.190.62.39 HTTP 467 GET /css/images/banner.svg HTTP/1.1
15469 42.523470 192.168.0.110 146.190.62.39 HTTP 482 GET /css/images/header-major-on-light.svg HTTP/1.1
15506 42.751501 146.190.62.39 192.168.0.110 HTTP/X. 1381 HTTP/1.1 200 OK
15511 42.768617 146.190.62.39 192.168.0.110 HTTP/X. 1331 HTTP/1.1 200 OK
15520 42.772067 192.168.0.110 146.190.62.39 HTTP 481 GET /css/images/header-major-on-dark.svg HTTP/1.1
15558 42.881824 192.168.0.110 146.190.62.39 HTTP 440 GET /favicon.ico HTTP/1.1
15678 43.038395 146.190.62.39 192.168.0.110 HTTP/X. 1337 HTTP/1.1 200 OK
15704 43.131057 146.190.62.39 192.168.0.110 HTTP 967 HTTP/1.1 200 OK (image/x-icon)
16410 75.703886 23.204.89.56 192.168.0.110 HTTP 564 [TCP Retransmission] HTTP/1.0 408 Request Time-out (text/html)
19069 122.053143 192.168.0.110 85.195.179.37 HTTP 303 GET /d/ndownload/update/others/2026/02/44699321_004610041f2cd2983510ea805fa8ddcc412a2fc4.cab HTTP/1.1
19076 122.360018 85.195.179.37 192.168.0.110 HTTP 749 HTTP/1.1 200 OK (application/vnd.ms-cab-compressed)
19078 122.368935 192.168.0.110 85.195.179.37 HTTP 303 GET /d/ndownload/update/others/2026/02/44699320_087bda2c210debad2477bd16d9a42091e80bf9e.cab HTTP/1.1
19084 122.666781 85.195.179.37 192.168.0.110 HTTP 743 HTTP/1.1 200 OK (application/vnd.ms-cab-compressed)
19088 122.673119 192.168.0.110 85.195.179.37 HTTP 303 GET /d/ndownload/update/others/2026/02/44699319_10a0b60f3908cd43535301f0561e00bc34836083.cab HTTP/1.1
19093 122.974616 85.195.179.37 192.168.0.110 HTTP 743 HTTP/1.1 200 OK (application/vnd.ms-cab-compressed)
19095 122.978602 192.168.0.110 85.195.179.37 HTTP 303 GET /d/ndownload/update/others/2026/02/44699318_33903c7001aa3457584e764940df4ff4d1efb5b0.cab HTTP/1.1
19101 123.282474 85.195.179.37 192.168.0.110 HTTP 751 HTTP/1.1 200 OK (application/vnd.ms-cab-compressed)
19103 123.284574 192.168.0.110 85.195.179.37 HTTP 303 GET /d/ndownload/update/others/2026/02/44699317_ff07ab7f66cf9a9a725018f02200519271c69555.cab HTTP/1.1

# Frame 15558: Packet, 440 bytes on wire (3520 bits), 440 bytes captured (3520 bits) on interface \Device\NPF_{E0EE93B-9C08-4742-A6E5-...}
# Ethernet II, Src: Intel_21:0e:c6 (70:a8:d3:21:0e:c6), Dst: TPLink_6a:5a:80 (14:eb:b6:6a:5a:80)
# Destination: TPLink_6a:5a:80 (14:eb:b6:6a:5a:80)
# Source: Intel_21:0e:c6 (70:a8:d3:21:0e:c6)
# Type: IPv4 (0x0800)
# Internet Protocol Version 4, Src: 192.168.0.110, Dst: 146.190.62.39
# Transmission Control Protocol, Src Port: 15892, Dst Port: 80, Seq: 765, Ack: 2246, Len: 386
# Hypertext Transfer Protocol
# GET /favicon.ico HTTP/1.1\r\n
# Host: httpforever.com\r\n
# User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/144.0.0.0 Safari/537.36\r\n
# Accept: image/avif,image/webp,image/apng,image/svg+xml,image/*,*/*;q=0.8\r\n
# Sec-GPC: 1\r\n
# Accept-Language: en-US,en;q=0.7\r\n
# Referer: http://httpforever.com/\r\n
# Accept-Encoding: gzip, deflate\r\n
# [Response in frame 15704]
# [Full request URI: http://httpforever.com/favicon.ico]
```

This capture shows HTTP GET requests sent from the local device to the web server, along with successful HTTP responses.

Part 2: Analyzing TCP/IP Traffic

Task 1: Filter TCP packets



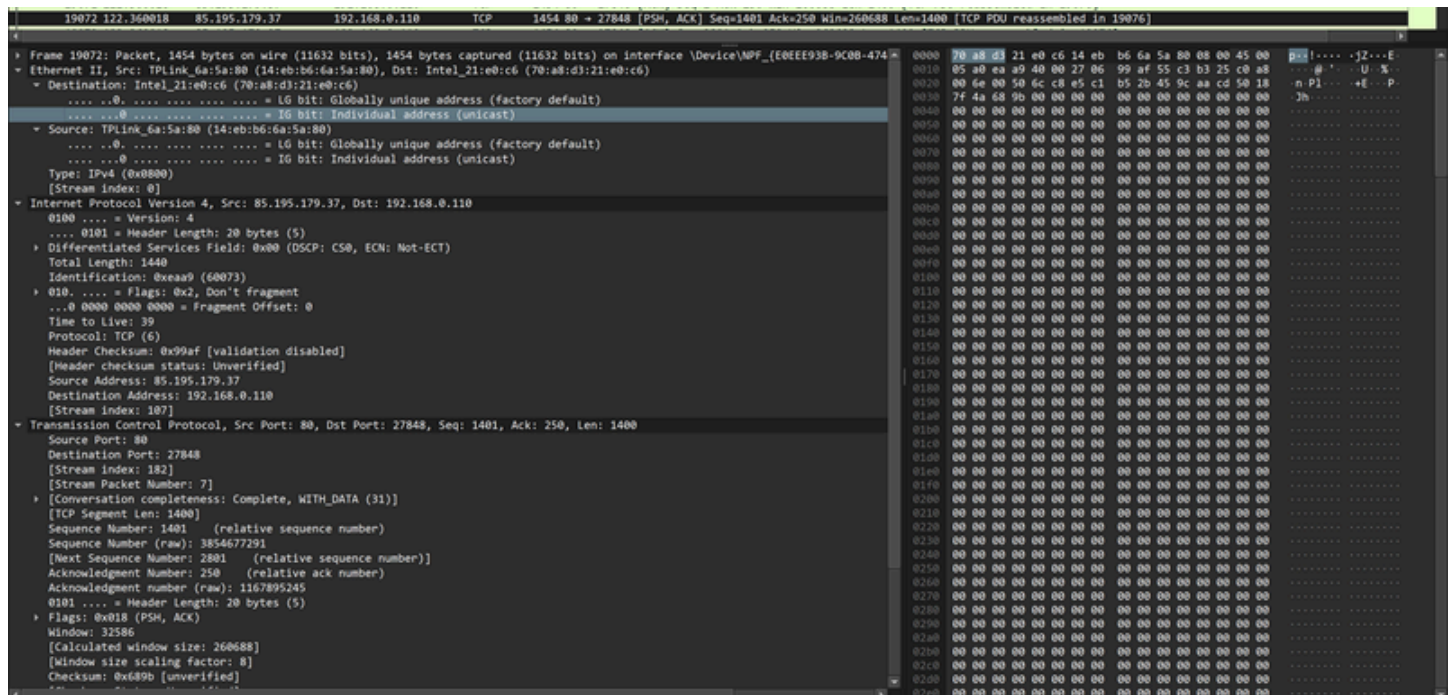
Task 2: Analyze TCP handshake and investigate Data Transfer and Termination

step 1 :

No.	Time	Source	Destination	Protocol	Length	Info
19065	121.805768	192.168.0.110	85.195.179.37	TCP	66	27848 → 80 [SYN] Seq=0 Win=65535 Len=0 MSS=1460 WS=256 SACK_PERM
19067	122.051964	85.195.179.37	192.168.0.110	TCP	66	80 → 27848 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1400 SACK_PERM WS=8
19068	122.052234	192.168.0.110	85.195.179.37	TCP	54	27848 → 80 [ACK] Seq=1 Ack=1 Win=65280 Len=0

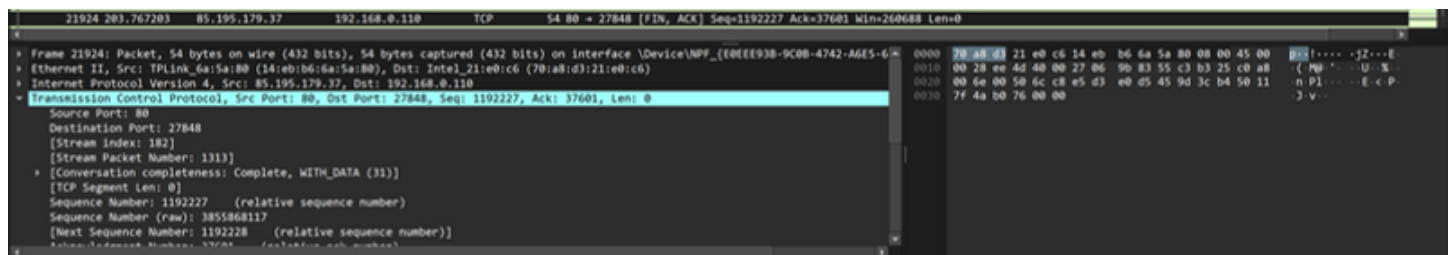
This screenshot shows the TCP three-way handshake. The connection starts with a SYN packet sent from the client to the server. The server responds with a SYN-ACK packet, and the client replies with an ACK packet. This process establishes the TCP connection before data transfer. This screenshot shows the TCP stream between the client and the server, displaying the HTTP request and the corresponding server response.

step 3 :



This screenshot shows TCP data transfer between the client and the server. Packets with PSH and ACK flags indicate that application data is being transmitted after the TCP connection was successfully established.

step 4 :



This screenshot shows the TCP connection termination process. The server initiates the termination by sending a FIN packet, and the client acknowledges it using an ACK packet. This indicates a normal and graceful termination of the TCP connection.

Part 3: Capturing and Analyzing UDP Traffic


```
No.    Time           Source                Destination           Protocol Length Info
10 0.448221 192.168.0.110        104.18.32.47         UDP 1292 59567 → 443 Len=1250
11 0.448251 192.168.0.110        104.18.32.47         UDP 1292 59567 → 443 Len=1250
12 0.448283 192.168.0.110        104.18.32.47         UDP 202 59567 → 443 Len=160
13 0.448511 192.168.0.110        104.18.32.47         UDP 751 59567 → 443 Len=709
14 0.449936 104.18.32.47         192.168.0.110        UDP 203 443 → 59567 Len=161
16 0.456540 192.168.0.110        104.18.32.47         UDP 88 59567 → 443 Len=46
17 0.480385 104.18.32.47         192.168.0.110        UDP 66 443 → 59567 Len=24
18 0.483531 104.18.32.47         192.168.0.110        UDP 66 443 → 59567 Len=24
19 0.488202 104.18.32.47         192.168.0.110        UDP 66 443 → 59567 Len=24

Frame 10: Packet, 1292 bytes on wire (10336 bits), 1292 bytes captured (10336 bits) on interface \Device\NPF_{E0EE93B-9C08-4742-A6E...
Ethernet II, Src: Intel21e0:c6 (78:a8:d3:21:e0:c6), Dst: TPLink_6a:5a:80 (14:eb:b6:6a:5a:80)
  Destination: TPLink_6a:5a:80 (14:eb:b6:6a:5a:80)
    ....0. .... = IG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Source: Intel21e0:c6 (78:a8:d3:21:e0:c6)
    ....0. .... = IG bit: Globally unique address (factory default)
    ....0. .... = IG bit: Individual address (unicast)
  Type: IPv4 (0x0800)
  [Stream index: 0]
Internet Protocol Version 4, Src: 192.168.0.110, Dst: 104.18.32.47
  0100 .... = Version: 4
  ....0101 = Header Length: 20 bytes (5)
  Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 1278
  Identification: 0x3850 (14416)
  010. .... = Flags: 0x2, Don't fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: UDP (17)
  Header Checksum: 0x0000 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 192.168.0.110
  Destination Address: 104.18.32.47
  [Stream index: 0]
User Datagram Protocol, Src Port: 59567, Dst Port: 443
  Source Port: 59567
  Destination Port: 443
  Length: 1250
  Checksum: 0x4e53 [unverified]
  [Checksum status: Unverified]
  [Stream index: 0]
  [Stream Packet Number: 9]
  [Timestamps]
  UDP payload (1250 bytes)
Data (1250 bytes)
  Data [..]: 4201d49bb048651ab7ed465b008667d276b20208d1fcc1600506907c77477457e287601efac068416ad93481b3315d6c473fbd9ca67da1a12054
```

UDP traffic was captured using Wireshark while a streaming application was running. After applying the UDP filter, the source port, destination port, packet length, and payload were observed.

Compare the simplicity of UDP headers with TCP headers.

UDP uses a simple header and does not provide connection establishment, acknowledgments, or sequence numbers. Unlike TCP, UDP prioritizes speed over reliability.

Part 4: Comparing TCP and UDP by filling in the following tables. Save your work (e.g., in an MS Word document), and upload it to your online git repo.

Task 1: Fill in the following table and provide reasons

	TCP or UDP	Reasons
Reliability and Connection Establishment	TCP	TCP establishes a connection using a three-way handshake and ensures reliable data delivery using acknowledgments and retransmissions.
Data Integrity and Ordering	TCP	TCP guarantees data integrity and correct ordering using sequence numbers and acknowledgment numbers.

Task 2: Identify the use Cases and Performance of TCP and UDP.

	TCP	UDP
Use cases	Web browsing-File transfer	Video streaming- video calls
Performance	Slower due to connection setup and reliability mechanisms	Faster because there is no connection setup